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Electromyographic comparison of vastus medialis obliquus facilitatory versus vastus lateralis inhibitory kinesio taping in athletes with patellofemoral pain: A randomized clinical trial☆

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ABSTRACT

Background: Patellofemoral pain (PFP) is among the most common orthopedic complications afflicting active young people. Diminished coordination of the vastii, delayed activation of the vastus medialis obliquus (VMO), and decreased VMO-to-vastus lateralis (VL) activity ratio are well documented as underlying causes of PFP. This study compares the effects of VMO facilitatory kinesio tape (KT) versus VL inhibitory KT on electromyographic (EMG) activity of the vastii, balance, and pain in athletes with PFP.
Methods: In this single-blind randomized clinical trial, thirty-two female athletes with PFP (mean age 26.33 ± 5.93 years) were randomly assigned to VMO facilitatory KT (n = 16) or VL inhibitory KT (n = 16) groups. In the facilitatory group, a Y-shaped strip of KT at 25% of its available tension was attached from the origin of the VMO to its insertion and in the inhibitory group, an insertion-to-origin Y-shaped strip of KT at 15% of its available tension was applied on the VL. Pain intensity, dynamic balance, and EMG data were assessed respectively with a visual analogue scale, the modified Star Excursion Balance Test, and an EMG telemetry system, before and immediately after KT application.
Results: Pain intensity decreased and dynamic balance improved significantly after taping in both groups, and VMO: VL activity ratio increased significantly in the VL group. However, none of the parameters differed significantly between groups.
Conclusions: Both VMO facilitatory and VL inhibitory KT can improve pain and balance, while the inhibitory technique might be more effective in regulating the VMO to VL activity ratio in athletes with PFP.

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1. Introduction

Diffuse retropatellar or peripatellar pain, known as patellofemoral pain (PFP), is the most common orthopedic problem among active young people, afflicting about 26% of young athletes (Lake & Wofford 2011) and accounting for the chief complaint in 30% of patients referred to sport medicine clinics (LaBella 2004). The PFP is aggravated during physical activities that increase compressive forces on the patellofemoral joint such as squatting, prolonged sitting, kneeling, and stair negotiation (Gaitonde et al., 2019). Although the exact etiology of PFP is not yet known, some biomechanical and neuromotor risk factors have been proposed including lower limb malalignment, joint laxity, quadriceps weakness, and imbalance between the vastii muscles (Neal et al., 2019).

The role of quadriceps strength and the coordination among its heads in individuals with PFP has attracted considerable attention among researchers and clinicians. In contrast to the simultaneous activation of the vastus medialis obliquus (VMO) and vastus lateralis (VL) in healthy people, delayed activation of the VMO compared to the VL and reduced VMO:VL activity amplitude is widely reported in patients with PFP (Cowan et al. 2001a,b; Mirzaie et al., 2019). The impaired muscle balance between the vastii heads can

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result in the VMO weakness and/or the VL tightness and in turn, cause lateral patellar tilt (Sisk and Fredericson 2019). The consequent patellar maltracking may lead to patellofemoral cartilage damage and secondary osteoarthritis (Eijkenboom et al., 2020).

Common therapeutic measures suggested for PFP include electrotherapy (Talbot et al., 2020), biofeedback and neurofeedback training (Ahmadi et al., 2020; Alonazi et al., 2021), quadriceps strengthening exercises (Naidu and Kage, 2018), core muscles training (Foroughi et al., 2019), knee orthoses (Smith et al., 2015), dry needling (Ma et al., 2020), and taping techniques (Aghapour et al., 2017). Kinesio tape (KT) is a porous, elastic, latex-free tape invented by Kase commonly used in sports medicine to prevent or manage a variety of musculoskeletal disorders (Kase et al., 2003). Several mechanisms have been suggested for KT including enhanced joint and muscle sensory inputs, increased lymphatic system activity, improved blood circulation, and pain relief via endogenous analgesic mechanisms. It is also claimed that KT can improve proprioception by normalizing muscle tone, correcting patellar maltracking, and stimulating skin mechanoreceptors (Bischoff et al., 2018; Freedman et al., 2014).

Some researchers believe that KT can produce both facilitatory or inhibitory effects depending on the direction of application as well as the amount of tension applied. When applied from origin to insertion of weak muscles, KT can increase muscle contraction and facilitate muscle activity. In contrast, an insertion-to-origin application of KT on overused muscles is thought to induce muscle inhibition (Kase et al., 2003; Kuo and Huang 2013). Previous studies have reported that the direction of KT does not affect the strength or electromyographic (EMG) activity of quadriceps in healthy individuals (Choi and Lee 2018; Serrão et al., 2016; Vercelli et al., 2012). In the individuals with PFP, however, the direction of KT application may affect the disturbed balance in the onset time and amplitude of vastii activation. To the best of our knowledge, only Chen et al. have taken Kase's view regarding taping direction into account and have used KT to pull the VMO up and pull the VL down concomitantly in patients with PFP. Chen et al. reported improvements in vastii EMG parameters compared to the “placebo tape” and “no tape” groups, while no direct comparison was made between the two directions of KT (Chen et al., 2007).

On the basis of Kase's theory, it can be hypothesized that facilitatory and inhibitory KT should be applied over the VMO and VL, respectively, in patients with PFP in order to improve the vastii neuromuscular control and to correct the patellar maltracking. However, it is not clear whether which method can be more effective. The present study was designed to compare the effectiveness of VMO facilitatory versus VL inhibitory KT on EMG activity of the vastii muscles, dynamic balance, and pain intensity in athletes with PFP.

2. Material and methods

2.1. Study design

In this single-blind randomized clinical trial, 32 female athletes with unilateral PFP were recruited by convenience sampling from sports clubs and were randomly assigned to the VMO facilitatory KT (n = 16) or VL inhibitory KT (n = 16) group with an online randomizer program (www.randomizer.org). The sample size was calculated with Power SSC as 32 patients (16 individuals in each group) to detect a between-group difference of 11.91 ms for VL-VMO onset time based on the data obtained from a pilot study on 10 patients (power = 80%, $\alpha = 0.05$, $\mu1-\mu2 = 11.91$, and $\sigma = 12.02$). The study was approved by the local medical ethics committee in accordance with the standards of the Declaration of Helsinki, under protocol number ct-9373-7053, and prospectively registered at the Iranian Registry of Clinical Trials (http://www.irct.ir/; IRCT2014071515932N3).

2.2. Participants

The volunteers were volleyball and basketball players who did the sports for at least three sessions of 2 h per week, for over one year. The PFP was diagnosed by an expert physical therapist with over 10 years of experience in sports rehabilitation and was confirmed by an orthopedic surgeon. The inclusion criteria were aged 18–40 years; a positive patellar grind test; unilateral pain in their dominant limb for at least two months caused by at least two of the following activities: running, hopping, kneeling, squatting, prolonged sitting, and stair negotiation; pain level over three on a 0-to-10 point visual analogue scale (VAS) during squatting; and functional level of at least 11 on the Functional Index Questionnaire (FIQ) (Samani et al., 2020). The Persian version of the FIQ with an intraclass correlation coefficient = 0.84 and Cronbach's alpha = 0.79 was used in this study (Negahban et al., 2013).

Participants were excluded if they had any history of hip or low back problems, lower limb surgery, patellar dislocation or subluxation, meniscal and ligament injuries, knee cartilage lesions, secondary knee osteoarthritis, or structural deformities in the spine or lower extremities. Patients with neurological or metabolic diseases and individuals who had received physiotherapy during the last three months or had used analgesic drugs in the last 72 h prior to the trial were also excluded (Cowan et al., 2006; Kamali et al., 2018).

2.3. Procedures

After signing informed consent forms, the participants were randomized into the VMO facilitation and VL inhibition groups, and their demographic characteristics were recorded. The examinees got familiar with the interventions and assessments through verbal instructions and practical demonstration of the tests. The baseline values of EMG signals, pain intensity, and dynamic balance were measured, the KT was applied, and immediately all variables were reassessed. In order to avoid fatigue, a 2-min rest interval was considered between the EMG and balance assessments. All measurements were conducted by an assessor blinded to group assignments.

2.4. Outcome measures

2.4.1. Electromyographic data recording

The EMG activities of the VMO, VL, and soleus muscles during five heel-rise tasks were recorded with an ME6000 16-channel EMG telemetry system (Mega Electronics Ltd, Kuopio, Finland) and transmitted to a computer. The perturbation induced by the heel-rise task in a closed kinetic chain can challenge knee stability by activating the soleus muscle (Cowan et al. 2001a,b). Since the soleus is the prime mover of the heel-rise and is activated prior to other muscles in this task, its onset time was considered as zero and the EMG onset times of the VMO and VL muscles were determined relative to the soleus onset (Biabanimoghadam et al., 2016).

The skin was shaved, cleaned with alcohol, and abraded with fine sandpaper to reduce electrical impedance. Then the round, self-adhesive silver/silver chloride surface electrodes (Solidor® MSGST-06, Medico Electrodes International, Uttar Pradesh, India) with a center-to-center distance of 20 mm were placed over the target muscles according to SENIAM standards (Stegeman and Hermens, 2007), as follows:

VMO: 4 cm superior and 3 cm medial to the superomedial border of the patella, at an angle of 55° from the vertical.

VL: 10 cm superior and 6–8 cm lateral to the upper border of the

patella, at an angle of 150° from the vertical.

Soleus: midpoint of the posterior surface of the calf, medial to the triceps surae tendon.

Ground electrodes for the VMO, VL, and soleus muscles were placed on the tibial tubercle, fibular head, and lateral malleolus, respectively (Cowan et al., 2002; Janwantanakul and Gaogasigam 2005). The patients stood with their arms by their sides and their weight equally distributed on both feet and were asked to perform a heel-rise as quickly and intensely as possible in response to the sound of a digital countdown timer (XNote Stopwatch, version 1.5c, dnSoft Research Group, Cheboksary, Russia). Participants were instructed to avoid balancing on their toes or performing undesired movements during the heel-rise tasks (Cowan et al., 2002). The EMG data were sampled at 1000 Hz, preamplified, band-pass filtered at 20–450 Hz, converted through a 14-bit A-D converter, and averaged over five repetitions with 30-sec rest intervals (Cowan et al., 2002; Janwantanakul and Gaogasigam 2005; Biabanimoghadam et al., 2016).

The EMG signals were stored as ASCII files and were analyzed with MATLAB 7.13 (MathWorks Inc., Natick, MA, USA) by an EMG data analyst blinded to the treatment allocation. After visual checks for any noise or motion artefacts, the EMG signals were fully rectified and then low-pass filtered at 50 Hz. The time point at which the amplitude of muscle activity exceeded the baseline value by three standard deviations for at least 25 ms was considered as the muscle EMG onset time. The vastii onsets were quantified by subtracting soleus EMG onset from the vastii onset latencies, then averaged over five trials. The root mean square of EMG amplitude was recorded for 200 ms after the onset of EMG activity with a 15-ms time window, and then was normalized to the peak amplitude (Straub and Cipriani, 2012). Normalization of the muscle activation level by the peak level is a standard method of EMG data normalization, representing the EMG signals during the executed task as a percentage of the peak muscle activity occurred during that task (Halaki and Ginn, 2012).

2.4.2. Dynamic balance

To assess dynamic balance, we used a modified version of the Star Excursion Balance Test (mSEBT) with excursion only in the anterior, anteromedial, and anterolateral directions (Gribble et al., 2012). According to Earl and Hertel, excursions in the anterior directions require higher levels of VMO activity and consequently, may be affected more than other directions in people with PFP (Earl & Hertel 2001). After the familiarization process, three trials were performed in each direction of the mSEBT and the mean value of three trials was divided by the patient's leg length and then multiplied by 100 to obtain the normalized reach distances as percentages (Aminaka and Gribble 2008). The mean of normalized reach distances in three directions was then recorded as the composite reach distance in order to evaluate one's general function in the mSEBT (Plisky et al., 2006).

2.4.3. Pain intensity

Pain intensity was assessed through a 0-to-10 VAS during performing the mSEBT which entailed unilateral mini-squats while standing on the affected leg (Aminaka and Gribble 2008; Miller et al., 2013). The VAS is a valid and sensitive tool in which 0 indicates “no pain” and 10 shows “the worst perceived pain” (Carlsson 1983).

2.5. Interventions

2.5.1. VMO facilitatory KT

After skin preparation, the VMO was taped with a Y-shaped strip of KT (TS Co., Seoul, South Korea) at 25% of its available tension. To induce a facilitatory effect, the KT was attached from the origin to the insertion (Kase 2003; Słupik et al., 2007) (Fig. 1).

2.5.2. VL inhibitory KT

The skin was prepared and a Y-shaped strip of KT was applied at 15% of its available tension from the insertion to the origin of the muscle (Kase 2003) (Fig. 2).

2.6. Statistical analysis

The data were analyzed with Statistical Package for Social Sciences (SPSS) software, version 21.0 (IBM Corp., Armonk, NY, USA), and P-values < 0.05 were considered statistically significant. The normal distribution of the data was confirmed by the one-sample Kolmogorov–Smirnov test ($P < 0.05$), so paired and independent t-tests were used for within-group and between-group comparisons, respectively.

3. Results

The flowchart of this study is shown in Fig. 3. As seen in Table 1, there were no significant differences between the two groups in any demographic or clinical characteristics at baseline. The pre-post assessments of EMG parameters revealed that neither facilitatory nor inhibitory KT was associated with a statistically significant change in the onset time of the vastii or delayed onset of the VMO relative to the VL. The VMO amplitude increased and the VL amplitude decreased significantly following KT application, and the VMO: VL activity ratio showed a significant increase in the VL group (Table 2). Pain intensity decreased and composite reach distance of the balance test increased significantly after both taping methods.

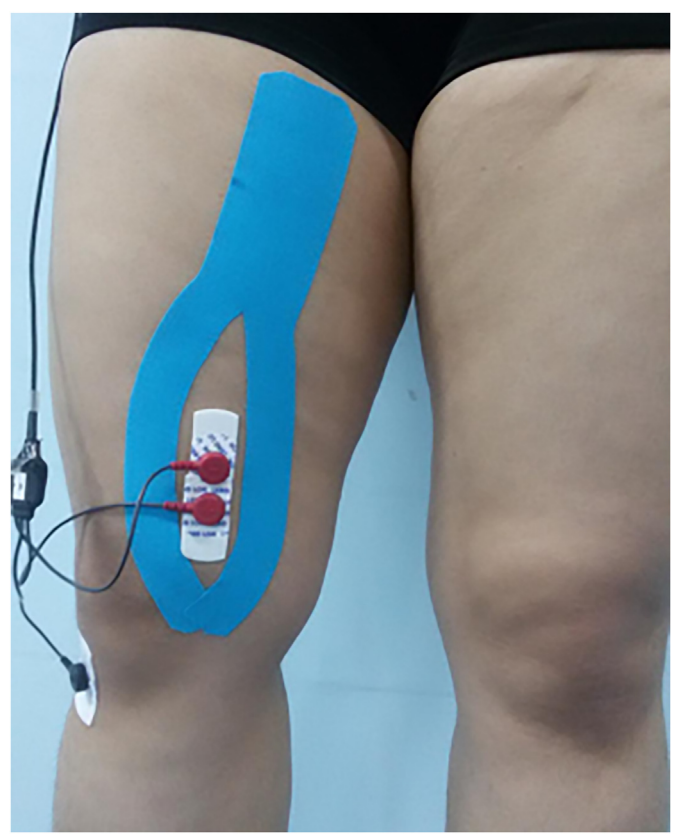

Fig. 1. Vastus medialis obliquus facilitatory kinesio taping technique.

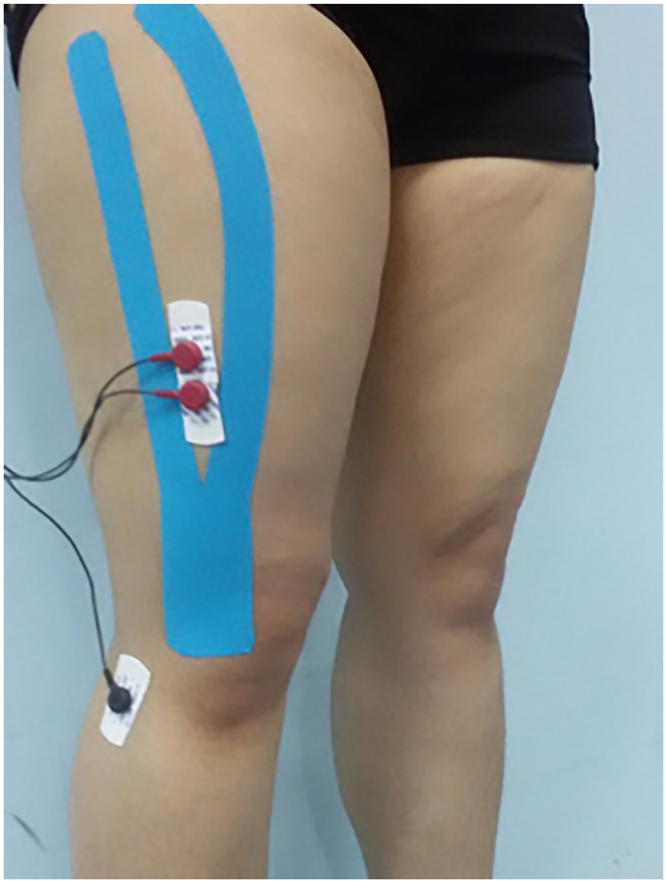

Fig. 2. Vastus lateralis inhibitory kinesio taping technique.

However, significant changes were reported in all three directions in the VMO group and only in the anterolateral direction in the VL group (Table 2). Between-group comparisons, however, showed no statistically significant differences for any of the outcome measures (Table 3).

4. Discussion

The results of this study demonstrated that the application of KT could significantly reduce pain and improve balance performance and EMG activity of the vastii in patients with PFP. The VMO activity amplitude increased significantly in the VMO facilitatory group, and VL activity amplitude decreased significantly in the VL inhibitory group, and these results were consistent with Kase's view regarding the effects of different directions of KT application (Kase 2003; Słupik et al., 2007). In the present study, KT was applied from the origin to the insertion of the VMO, in alignment with the muscle fibers. One conceivable mechanism for the effect of facilitatory KT is its pull on the fascia and recoil in the direction of muscle contraction, which can improve muscle contraction (Kuo and Huang 2013). Moreover, increased afferent inputs to the dorsal horn of the spine can reduce the pain-related inhibition of alpha motor neuron excitability, resulting in more efficient VMO contraction (Herrington 2001). In contrast, inhibitory KT can pull the muscle fibers in the opposite direction of muscle contraction and induce an inhibitory effect by stretching the Golgi tendon organs at the distal end of the muscle (Kuo and Huang 2013).

However, the VMO: VL activity ratio which is a predisposing factor for the development of PFP may be of more clinical importance than the VMO or VL activity alone in designing the rehabilitation programs for these patients (Ng et al., 2008). In the present study the VMO: VL activity ratio was significantly altered after the application of inhibitory KT over the VL, whereas no significant

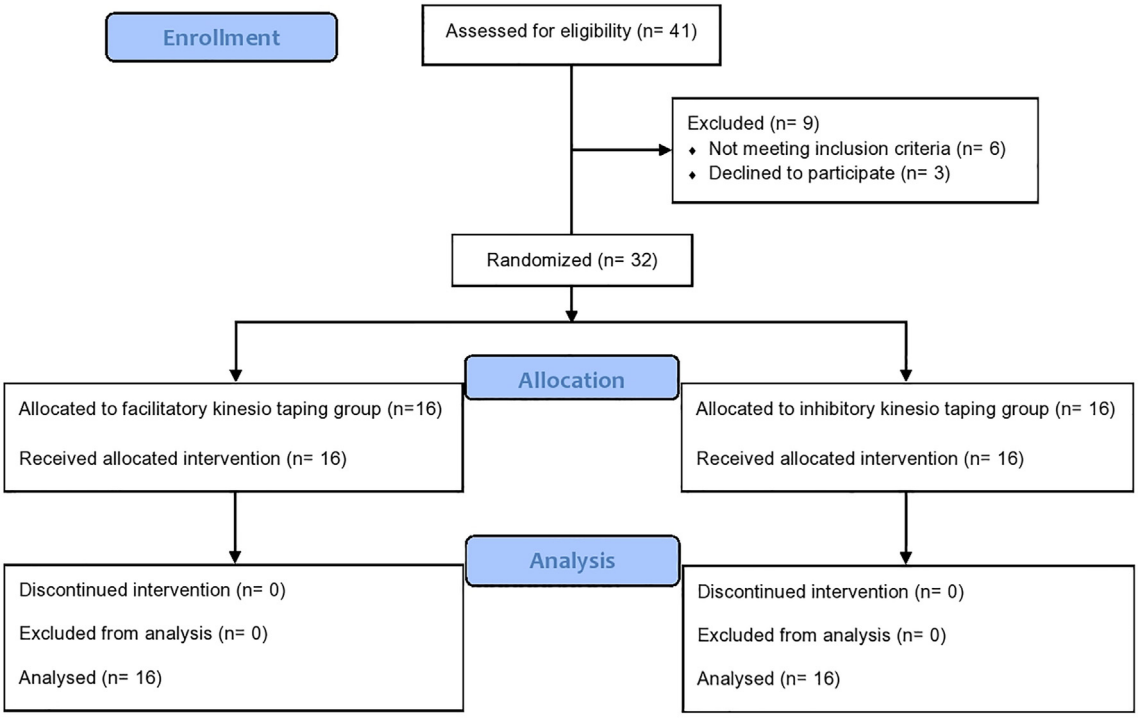

Fig. 3. Flow diagram showing participants in each phase of the study.

Table 1
Demographic and baseline characteristics of the participants.

Variable	VMO facilitatory KT (n = 16)	VL inhibitory KT (n = 16)	*P*-value
Age (years)	27.37 ± 6.40	25.25 ± 4.98	0.304
Height (cm)	160.75 ± 5.23	161.81 ± 4.51	0.544
Weight (kg)	59.12 ± 8.67	56.93 ± 7.61	0.454
VAS (0–10)	4.65 ± 1.32	4.81 ± 1.27	0.737
FIQ (0–16)	12.06 ± 1.18	11.87 ± .88	0.615

Abbreviations: VMO = Vastus medialis obliquus, VL = Vastus lateralis, KT = Kinesio tape, VAS = Visual analogue scale, FIQ = Functional Index Questionnaire.
Note: Values are mean ± SD.

Table 2
Within-group comparisons of pain intensity, dynamic balance and EMG data in vastii muscles before and after the interventions.

Outcome measures	VMO Facilitatory KT (n = 16)			VL Inhibitory KT (n = 16)		
	Pre	**Post**	***P*-value**	**Pre**	**Post**	*P*-value
VAS (0–10) mSEBT (cm)	2.37 ± 1.70	1.37 ± 1.31	0.021*	2.25 ± 2.11	1.43 ± 1.54	0.027*
Anterior	82.49 ± 13.94	89.58 ± 14.07	0.001*	80.57 ± 10.27	83.87 ± 12.32	0.078
Anteromedial	84.44 ± 14.51	92.73 ± 18.73	0.002*	83.80 ± 11.31	87.18 ± 11.83	0.072
Anterolateral	80.36 ± 15.88	86.93 ± 16.32	0.013*	77.88 ± 11.80	83.75 ± 12.16	0.002*
Composite	82.43 ± 14.27	89.75 ± 16.06	0.001*	80.75 ± 10.98	84.93 ± 11.77	0.012*
EMG data						
VMO Onset time (ms)	−143.15 ± 195.95	−127.96 ± 179.90	0.563	−157.09 ± 216.11	−110.18 ± 265.00	0.322
VL Onset time (ms)	−146.89 ± 197.21	−122.86 ± 191.21	0.375	−164.50 ± 224.79	−98.03 ± 281.58	0.218
VL-VMO Onset time (ms)	−3.73 ± 27.63	5.10 ± 32.09	0.304	−7.40 ± 33.91	12.15 ± 23.66	0.068
VMO Amplitude (μV)	33.17 ± 17.05	39.56 ± 19.08	0.024*	51.28 ± 12.55	49.84 ± 13.54	0.660
VL Amplitude (μV)	36.05 ± 16.61	38.40 ± 9.68	0.576	53.48 ± 23.08	47.02 ± 26.45	0.023*
VMO: VL Amplitude ratio	0.93 ± 0.20	1.09 ± 0.60	0.280	1.05 ± 0.36	1.25 ± 0.58	0.027*

Abbreviations: VMO = Vastus medialis obliquus, VL = Vastus lateralis, KT = Kinesio tape, VAS = Visual analogue scale, mSEBT = Modified Star Excursion Balance Test.
Note: Values are reported as mean ± standard deviation. Values marked with asterisk indicate a significant pre–post-treatment change.

Table 3
Between-group comparisons of mean changes of pain intensity, dynamic balance and EMG data in vastii muscles.

Outcome measures	VMO Facilitatory KT (n = 16)	VL Inhibitory KT (n = 16)	*P*-value
VAS (0–10) mSEBT (cm)	−1.00 ± 1.54	−0.81 ± 1.32	0.716
Anterior	7.09 ± 7.08	3.29 ± 6.96	0.137
Anteromedial	8.29 ± 8.98	3.37 ± 6.97	0.095
Anterolateral	6.57 ± 9.26	5.87 ± 6.36	0.805
Composite	7.31 ± 7.38	4.17 ± 5.84	0.193
EMG Data			
VL-VMO Onset Time (ms)	8.82 ± 33.21	19.55 ± 39.79	0.414
VMO: VL Amplitude Ratio	0.15 ± 0.56	0.20 ± 0.33	0.780

Abbreviations: VMO = Vastus medialis obliquus, VL = Vastus lateralis, KT = Kinesio tape, VAS = Visual analogue scale, mSEBT = Modified Star Excursion Balance Test.
Note: Values are reported as mean change ± standard deviation.

change was seen in the VMO facilitatory KT group. This finding can be explained by Janda's theory of muscle imbalance suggesting that when a group of muscles is affected by weakness or lack of tone, their antagonists are shortened or hypertonic, reciprocally inhibiting agonist muscles tone (Janda et al., 1996). In order to address muscle imbalances, Janda believed that priority should be given to overactive and hypertonic muscles with the aim of increasing tone in the weakened and inhibited muscles (Janda et al., 1996). In patients with PFP, therefore, it can be deduced that taping the hypertonic VL appears to be more effective than taping the weakened and inhibited VMO.

Regarding knee pain, both KT methods significantly decreased pain intensity with no significant difference between the two methods. The analgesic effects of KT can be attributed to various biomechanical and neuromuscular mechanisms including enhanced neural inputs (Akbaş et al., 2011), increased lymph and blood circulation, and blockage of nociceptive inputs through the "gate theory of pain" mechanism (Aytar et al., 2011). Moreover, pain relief induced by KT can lead to greater VMO activation and may help correct the patellar position within the trochlear groove. These effects in turn reduce pressure on the articular cartilage (Chen et al., 2007) and distribute the shear and compressive forces on the patellofemoral joint more evenly (Sikorski et al., 1979). In previous studies, some researchers applied KT around the patella and attributed PFP relief to the biomechanical correction of patellar alignment (Aminaka and Gribble 2008), while others taped the vastii and related pain mitigation to neurological suppression due to the stimulation of cutaneous mechanoreceptors (Aghapour et al., 2017; Shakeri et al. 2019). Kellish et al. also applied KT over the patella, VMO, and VL of patients with PFP to induce both mechanical and neuromuscular effects simultaneously, and reported significant pain relief immediately following KT application (Kellish et al., 2020). Considering the mechanisms suggested for the efficacy of KT in pain alleviation, it appears that the correct application of KT can diminish pain intensity in individuals with PFP regardless of the application direction. Sisk and Fredericson reviewed the common adjunctive treatments of PFP including taping and bracing and concluded that taping (both McConnell and KT) can considerably

reduce pain levels compared to no taping (Sisk & Fredericson 2020). Recent guidelines also recommend taping as a complementary treatment that can improve pain and function in PFP patients in the short term, while emphasizing its combination with exercise therapy as the gold standard treatment of PFP (Collins et al., 2018).

The composite mSEBT score, as a measure of functional balance, increased significantly in both groups following KT. Previous studies have reported that impaired the knee joint proprioception, which is an integral contributor of sensory inputs, due to PFP can afflict the sufferer's balance ability (Akseki et al., 2008; Baker et al., 2002). The improvement in our participants' balance performance may be due to increased proprioceptive inputs from the knee musculature, cutaneous mechanoreceptors, and joint proprioception induced by KT (Akbaş et al., 2011). Decreased pain intensity and kinesiophobia induced by KT are the other possible mechanisms that allow patients to bear weight on the affected leg more efficiently, thus improving their balance performance (Kurt et al., 2016). Our findings are in line with the study by Mostamand et al. who applied KT to induce medial gliding over the patella of PFP patients and reported significant improvement in pain and postural balance assessed by Y-balance test in anterior, posteromedial, and posterolateral directions. However, their findings may not be directly comparable with ours due to different taping techniques and various reach directions of balance tests (Mostamand et al., 2020). Earl and Hertel reported the VMO to be more activated in anterior-directed excursions of the SEBT than the other directions and also more activated compared to the VL in all directions of the SEBT (Earl & Hertel 2001). These findings imply that the VMO has a greater impact on the patient's anterior reach performance in the SEBT (i.e., anterior, anteromedial, and anterolateral reaching), and support our observation of significant improvement in all three directions in the VMO facilitatory group, whereas in the VL inhibitory group improvement was seen only in the anterolateral direction.

4.1. Study limitations

The present study only evaluated the immediate effects of KT and lacked a placebo taping or control group. Moreover, our samples were only female athletes, hindering the generalizability of the findings to all patients with PFP. Future studies with longer follow-up including other groups (e.g., placebo tape, no-tape, and simultaneous application of VMO and VL KT) are suggested to investigate the persistence of KT effects, rule out its placebo effect, and compare the combination of two techniques to each separately.

5. Conclusion

This study revealed that the VMO facilitatory and VL inhibitory KT can reduce pain, improve dynamic balance in female athletes with PFP. Although both techniques could significantly change the electrical activity of the muscles undergone KT, the inhibition of VL seems more effective to improve the VMO to VL activity ratio.

5.1. Clinical relevance

- Delayed onset of vastus medialis obliquus and decreased vastus medialis obliquus-to-vastus lateralis activity ratio is common in patients with patellofemoral pain.
- Lateral patellar tilt due to changes in vastii activities can reduce the force production ability of the vastus medialis obliquus.
- Facilitation of the vastus medialis obliquus or inhibition of the vastus lateralis by kinesio taping may address the patellar maltracking in patients with patellofemoral pain.
- Inhibitory kinesio taping over the vastus lateralis seems more effective than the facilitatory kinesio taping over the vastus medialis obliquus in improving the vastii activity ratio.

Ethical approval

Shiraz University of Medical Sciences Ethics Committee (Reference number: ct-9373-7053).

Clinical trial registry number (irct.ir)

IRCT2014071515932N3.

CRediT authorship contribution statement

Ehsan Sinaei: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Vahideh Foroozantabar:** Conceptualization, Data curation, Investigation, Writing – original draft. **Amin Kordi Yoosefinejad:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Sobhan Sobhani:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Alireza Motealleh:** Conceptualization, Methodology, Supervision, Formal analysis, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no conflict of interest.

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